



Figure 1: Popular deep learning libraries

Deep learning frameworks/libraries

There are many popular libraries that can be used to perform deep learning tasks. Some of them are listed below:

- Caffe
- Theano
- Torch
- DeepLearning4J
- Mocha

A detailed comparison of deep learning frameworks is available at <https://deeplearning4j.org/compare-dl4j-torch7-pylearn>.

DeepLearning4j (DL4J)

This article explores the DeepLearning4J (DL4J) library. DL4J has been developed in Java and is targeted at Java Virtual Machine (JVM). An interesting feature of DeepLearning4J is the ability to build fast prototypes. The attributes of DL4J are listed below.

- **Distributed:** The training process can be accelerated because DL4J is distributed by nature. It can harness multiple GPUs, and the performance is on par with other major libraries such as Caffe.
- **Open Source:** DL4J is a production quality open source library available for performing deep learning tasks. The active community of developers keeps DL4J fresh.
- **Interoperability:** DL4J is written in Java, and hence all the JVM based languages such as Scala, Clojure and Kotlin are compatible. Interoperability with other major frameworks such as Caffe and Torch can be achieved through Keras.

DL4J features

The major features of DL4J are:

- Java, Scala and Python APIs
- Parallel training through iterative reduce
- Scalable with Hadoop
- Distributed CPU and GPU support

DL4J incorporates both a distributed, multi-threaded deep learning framework and a single-threaded deep learning framework. Another important feature of DL4J



Figure 2: DL4J features

is that it is the first deep learning framework adopted for a microservice architecture.

Prerequisites

The prerequisites to start development with DL4J are listed below:

- **Java 1.7 or above** (DL4J supports only the 64-bit versions).
- **Apache Maven:** This is a dependency management and build tool. The Maven version in your system can be checked with the following command:

```
mvn --version
```

If you are new to Maven, an excellent 'getting started' guide (Maven in Five Minutes) is available at <http://maven.apache.org/guides/getting-started/maven-in-five-minutes.html>

- The official document recommends the use of IntelliJ IDE or Eclipse. The community edition of IntelliJ IDE can be downloaded from the official website.
- **Git:** Get the latest version with the following command:

```
$ git clone git://git.kernel.org/pub/scm/git/git.git
```

Using the DL4J examples

To download and use the examples from DL4J, use the following commands:

```
$ git clone https://github.com/deeplearning4j/dl4j-examples.git
$ cd dl4j-examples/
$ mvn clean install
```

1. Run IntelliJ. Choose the **Import Project** option. Select the folder 'dl4j-example'.
2. Select 'Import Project from external model' and make sure that Maven is chosen.
3. Follow the wizard's instructions. Click **Finish** to complete. It might take some time to get all the dependencies in the local system.